Dialect-Aware Automatic Speech Recognition

Research background and PhD vision

Satiyabooshan Murugaboopathy

Doctoral Researcher Position 1 · University of Tübingen

Background

ML researcher: low-resource, uncertainty-aware modeling and *systematic benchmarking* for real-world deployment.

- **Voice-agent startup (co-founder)** — production Whisper ASR (German), wav2vec 2.0 / HuBERT, CTC, on-device inference.
- **Linköping IAS** — uncertainty-aware pipelines for sparse, heterogeneous data.
- **Fraunhofer IMW/ISI** — NLP over noisy corpora; weak supervision.
- **Porsche & Volkswagen** — sequence, multimodal & small-sample learning.

Education

M.Sc. Computer Science, Uni Leipzig — thesis: systematic benchmarking of text classification.

B.Sc. Computer Science.

Why this project

Native **German** speaker; keen on *speech as a clinical biomarker*.

Research I — Understanding *why* models fail

Main tool: intervene in representation / calibration space, without new labels or retraining.

SHEAR — spurious shortcuts vs. sparse autoencoders
Decomposes frozen text embeddings into sparse features and *ablates* shortcut features with no group labels — works even when target and spurious attribute are **perfectly correlated**.

LUCID — statistically grounded self-correction
Diagnoses an LLM rater's *systematic bias* (doubly-robust DSL/PPI), feeds it back as text, and lets the rater re-grade itself — no retraining.

Thread: make cheap models *trustworthy* under scarcity.

Research II — Verifying representations under scarcity

Platonic Representations (poverty mapping)
Fuses **satellite vision + LLM/agent text** to predict wealth across ∼60k African DHS clusters; releases an open multimodal dataset. Fusion beats vision-only (R^2 0.77 vs. 0.63 OOS).

Agent Verifiers (counterfactual analysis)
Benchmarks whether agentic treatment-effect representations are *causally admissible* — guarding against retrieval and encoder leakage before they enter causal analysis.

Thread: benchmarking is itself a research contribution.

PhD Vision — Interpretability meets data-scarce ASR

Problem: SOTA speech recognition breaks on 1950–75 Swabian dialect — regional variation, spontaneous speech, almost no annotation.

- Adapt **self-supervised speech models** (wav2vec 2.0 / HuBERT / WavLM) to dialect — but *diagnose* the shift, don't just fine-tune.
- **SHEAR-style** sparse decomposition of speech features to find dialect / shortcut directions.
- **LUCID-style** calibrated uncertainty to route unsure segments to expert (human-in-the-loop) curation.

Deliverables: uncertainty-aware dialect ASR + IPA transcription, an open benchmark, and clinical validation on the TREND cohort.